IP Tunnels
The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

IP Tunneling

Jose L. Muñoz, Oscar Esparza, Juanjo Alins, Jorge Mata
Telematics Engineering
Universitat Politècnica de Catalunya (UPC)

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Outline

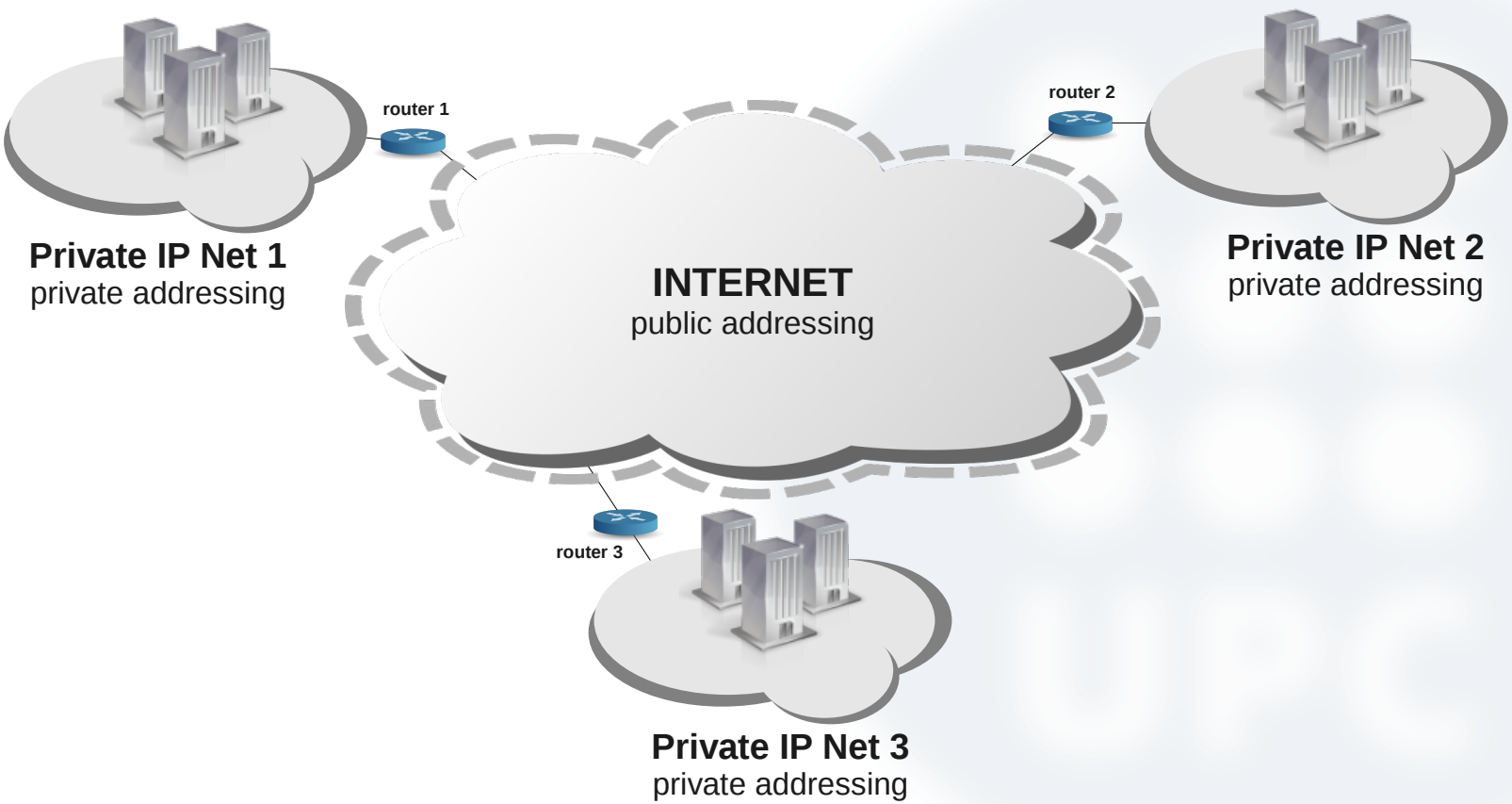
1 IP Tunnels
The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

The problem

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The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

Problem

Inter-connecting private IP networks through a public IP Network



Why is this a problem?

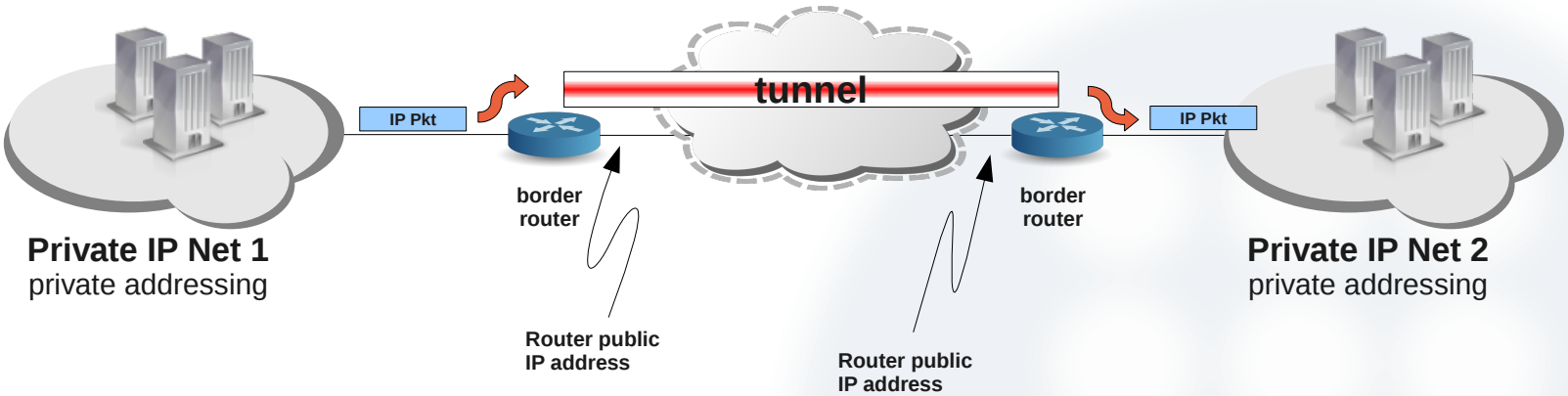
- Private IPs (RFC 1918) are used in internal networks.
  - A class: 10.0.0.0/8.
  - B class: 172.16.0.0/16-172.31.0.0/16.
  - C class: 192.168.0.0/16.
- But...
  - The public network does not route private IP addresses.
  - IP packets with a private destination IP address are silently discarded in public routers.
- This is a **routing** problem.

IP Tunnels
The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

The tunneling solution

Tunnel

A tunnel is essentially a network communications channel between two networks.



- Tunnels are used to transport another network protocol by encapsulation of its packets.
- They can be seen as Source Routes that circumvent conventional routing mechanisms.
- When the tunnel is built at IP layer it is named *IP tunnel*.
- The “outer” (public) router IP addresses identify the “endpoints” of the tunnel.

IP Tunnels
The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

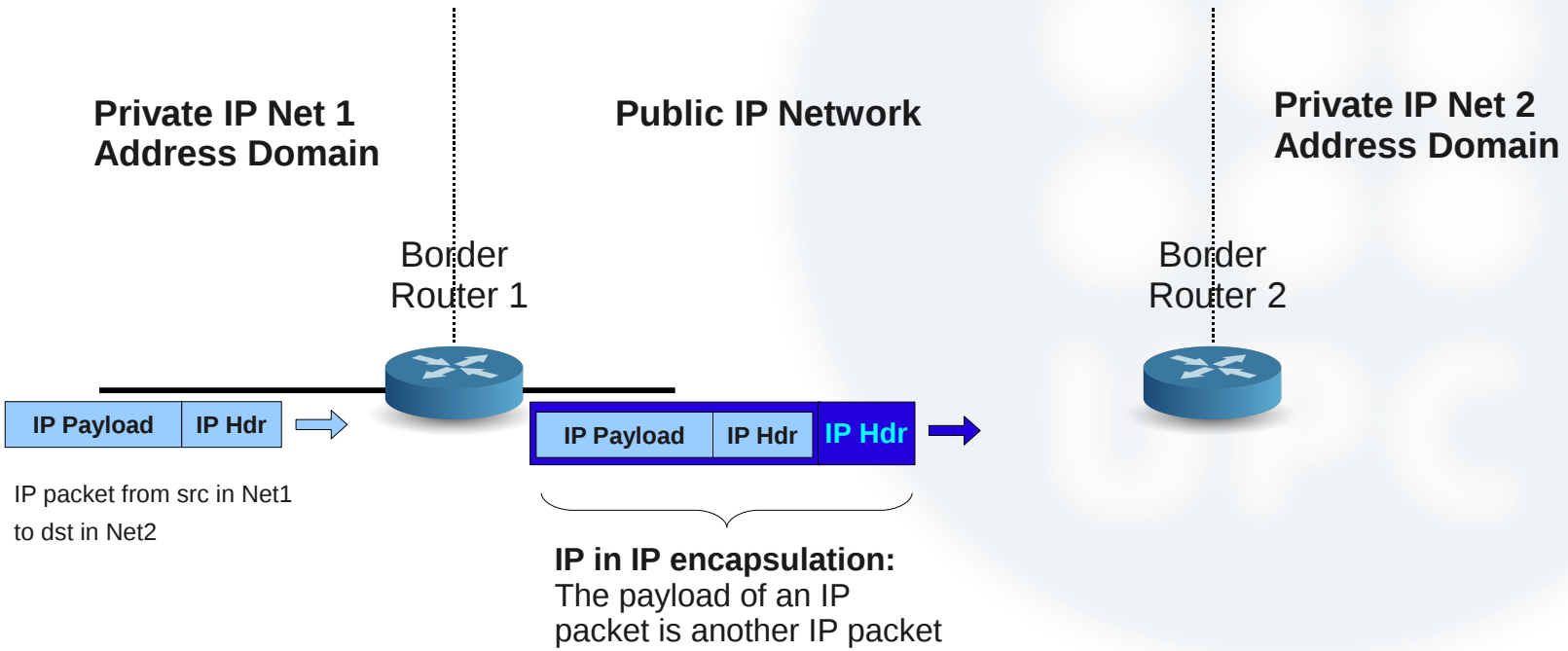
The IP tunnel

Every IP packet (including addressing information) is encapsulated within another packet, whose format is native to the public network.

There are several means of accomplishing this by using:

- IP as encapsulation protocol (ipencap, 6in4, 4in6).
- Other encapsulation protocols (IPsec, TLS, SSH, etc.).

A destination can be reached transparently without the source having to know topology specifics.



IP Tunnels
The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

The IP encap method (I)

Each border router creates a new network pseudo-device (associated to the outer IP address):

- When a packet is sent through this pseudo-device, it is encapsulated on a IP packet and sent to the other remote border router.
- When a packet is received through this pseudo-device, the payload is extracted and treated as an incoming real IP packet.

This network pseudo-device only accepts IP packets from the other border router.

IP Tunnels
The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

The IP encap method (II)

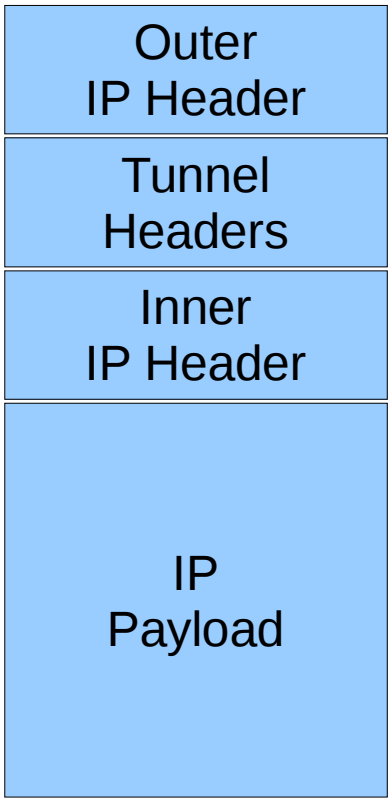
The routing table uses this network pseudo-device to know which packets will be sent through the IP tunnel.

```
Destination   Gateway   Genmask           Iface
...   ...
192.168.2.0   0.0.0.0   255.255.255.0     tunnel0
... ...
```

For example: any packet towards 192.168.2.0/24 will be sent through the `tunnel0` network pseudo-device, encapsulated in another IP packet and sent to the other remote border router.

IP Tunnels
The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

The outer IP packet



The outer IP packet presents this format:

- Outer IP Header:
  - IP source and destination addreses: these are the “endpoints” of the tunnel.
  - IP protocol field: states the type of encapsulation used.
- Tunnel Header: some encapsulation types define a specific tunnel header to introduce extra parameters.
- The encapsulated “inner” IP packet.

IP Tunnels
The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

Encap type: IP in IP I

- RFC 1853 Informational document (1995) : `IP in IP Tunneling`
- RFC 2003 Standard Track document (1996): `IP Encapsulation within IP`
- Encap can only encapsulate IPv4 unicast traffic.
- The protocol number associated to this encap type is 4.
- This encapsulation type does not define any Tunnel Header.

IP Tunnels
The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

Encap type: IP in IP II

- RFC 2003 addresses known problems due to encapsulation:
  - “Relaying” ICMP Messages from within the Tunnel:
    - The PMTU Discovery.
    - Time exceeded.
    - Other ICMP messages.
  - Tunnel Management:
    - Tunnel MTU Discovery.
    - Tunnel Congestion.

IP Tunnels
The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

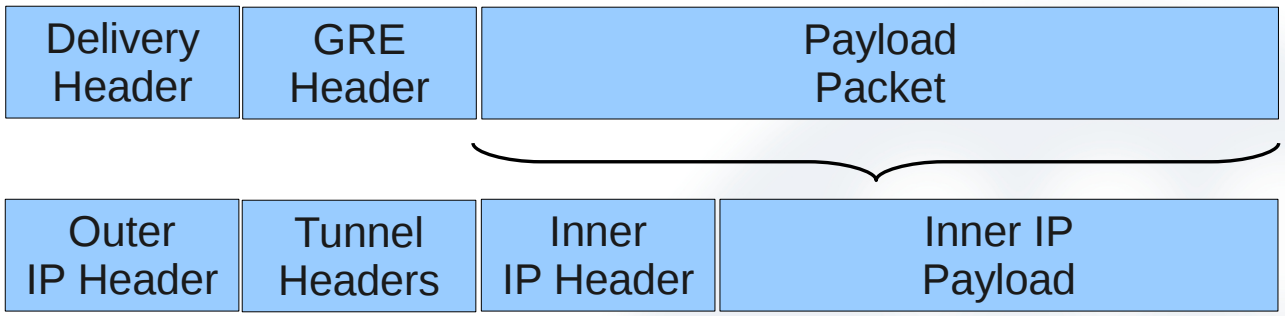
Encap type: GRE I

- Initially developed by Cisco Systems.
- RFC 1701 and RFC 1702 Informational documents (1994)
- RFC 2784 Standard Track document (2000): `Generic Routing Encapsulation (GRE)`
- GRE is more general than other proposals like encap.
- The protocol number associated to this GRE type is 47.

IP Tunnels
The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

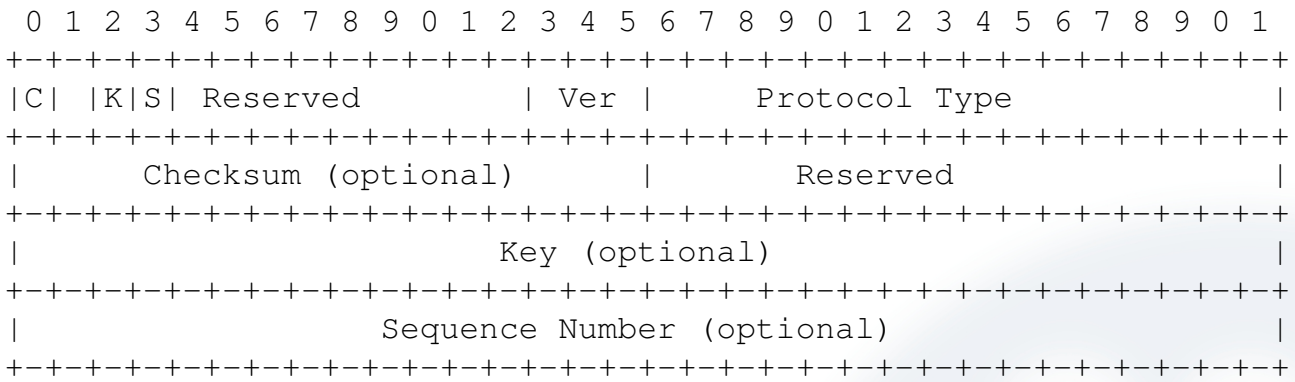
Encap type: GRE II

- A GRE encapsulated packet has the form:



- The RFC specification basically explains the structure of the GRE header.
- The GRE header is variable in length (4 to 16 bytes).
- There are optional fields.

GRE Header (version 0)



- **C**, **K**, and **S**: flags which are set if the checksum, key, and sequence number fields are used, respectively.
- **Ver**: version number (zero)
- **Protocol Type**: encapsulated protocol

Protocol Family	PTYPE
IPv4	0x0800
Ethertalk	0x809B
Novell IPX	0x8037
...	...

- **Checksum**: Packet checksum (optional)
- **Key**: Tunnel key (optional)
- **Sequence Number**: GRE sequence number (optional)

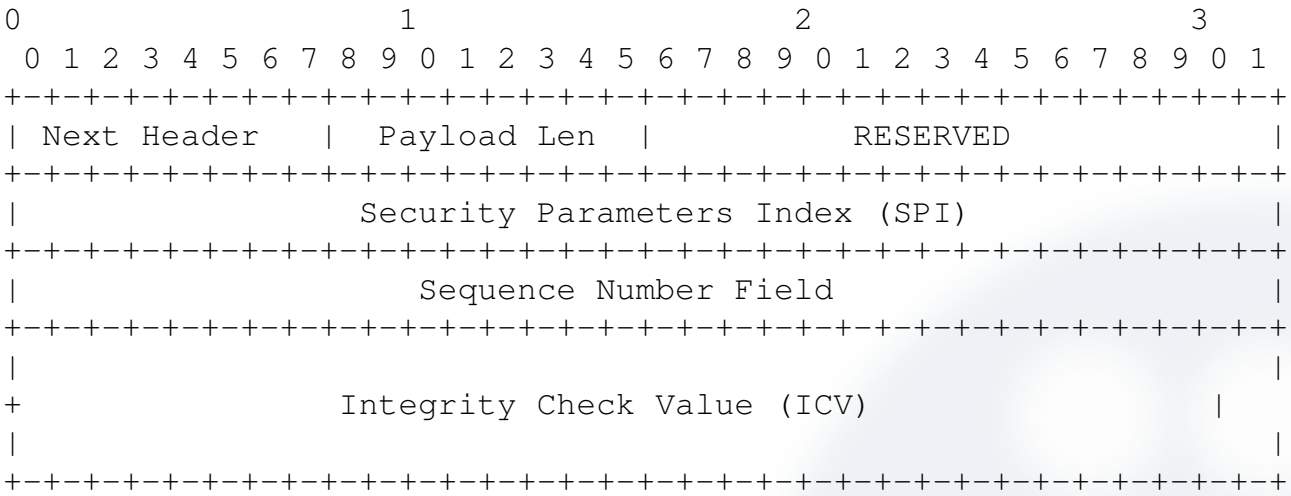
IP Tunnels
The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

Encap type: IPsec

- Internet Protocol Security (IPsec) can provide:
  - Authentication by using Authentication Headers (AH).
  - Encryption by using Encapsulating Security Payloads (ESP).
  - Mutual authentication at the beginning of the session and negotiation of cryptographic keys to be used during the session (using Security Associations (SA)).
- Defined in RFC 1825 through RFC 1829.
- Transport and tunnel modes.

IP Tunnels
The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

Authentication Header



- **Next Header**: type of the next header.
- **Payload Len**: length of this Authentication Header.
- **Security Parameters Index**: random value to identify the security association (SA) of the receiving party.
- **Sequence Number**: to prevent replay attacks.
- **Integrity Check Value**: to check the integrity of the packet.

IP Tunnels
The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

Configuration of tunnels I

- We will use the command `ip tunnel`
- The encapulating (or outer) address family is specified by the -f option. The default is IPv4.
  `ip tunnel show` - list tunnels
  `ip tunnel add` - add a new tunnel
  `ip tunnel change` - change an existing tunnel
  `ip tunnel delete` - destroy a tunnel
  `name NAME` - select the tunnel device name.
  `mode MODE` - set the tunnel mode. Available modes depend on the encapsulating address family.
  - Modes for IPv4 encapsulation available: `ipip, sit, isatap and gre`.
  - Modes for IPv6 encapsulation available: `ip6ip6, ipip6 and any`.

IP Tunnels
The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

Configuration of tunnels II

`remote ADDRESS` - set the remote endpoint of the tunnel.
`local ADDRESS` - set the fixed local address for tunneled packets. It must be an address on another interface of this host.
`ttl N` - set a fixed TTL N on tunneled packets. N is a number in the range 1–255. 0 is a special value meaning that packets inherit the TTL value. The default value for IPv4 tunnels is: inherit. The default value for IPv6 tunnels is: 64.

IP Tunnels
The Interconnection problem
The Tunneling Solution
IP tunnels
IP encapsulation types

Configuration of tunnels III

`dev NAME` - bind the tunnel to the device NAME so that tunneled packets will only be routed via this device and will not be able to escape to another device when the route to endpoint changes.
`nopmtudisc` - disable Path MTU Discovery on this tunnel. It is enabled by default. Note that a fixed ttl is incompatible with this option: tunnelling with a fixed ttl always makes pmtu discovery.